LAB EXPERIMENT RECORD

Set 1: Monthly Sales Data

Department	AI &DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To visualize monthly sales data using a line chart, bar chart, and scatter plot, and to build an interactive dashboard using R and Shiny.

Dataset: Monthly Sales

Month	Sales ($)
January	15000
February	18000
March	22000
April	20000
May	23000

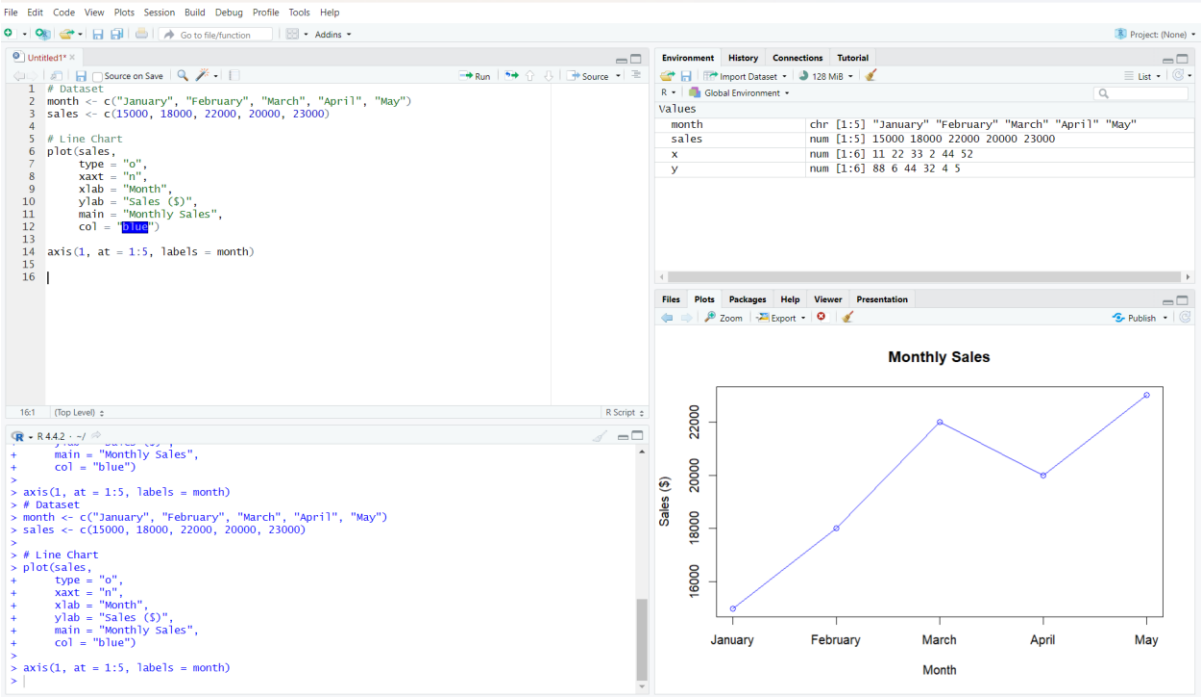
PROGRAM 1: Line Chart of Monthly Sales

Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

# Line Chart
plot(sales,
     type = "o",
     xaxt = "n",
     xlab = "Month",
     ylab = "Sales ($)",
     main = "Monthly Sales",
     col = "blue")

axis(1, at = 1:5, labels = month)
```



Output 1: Monthly Sales Line Chart

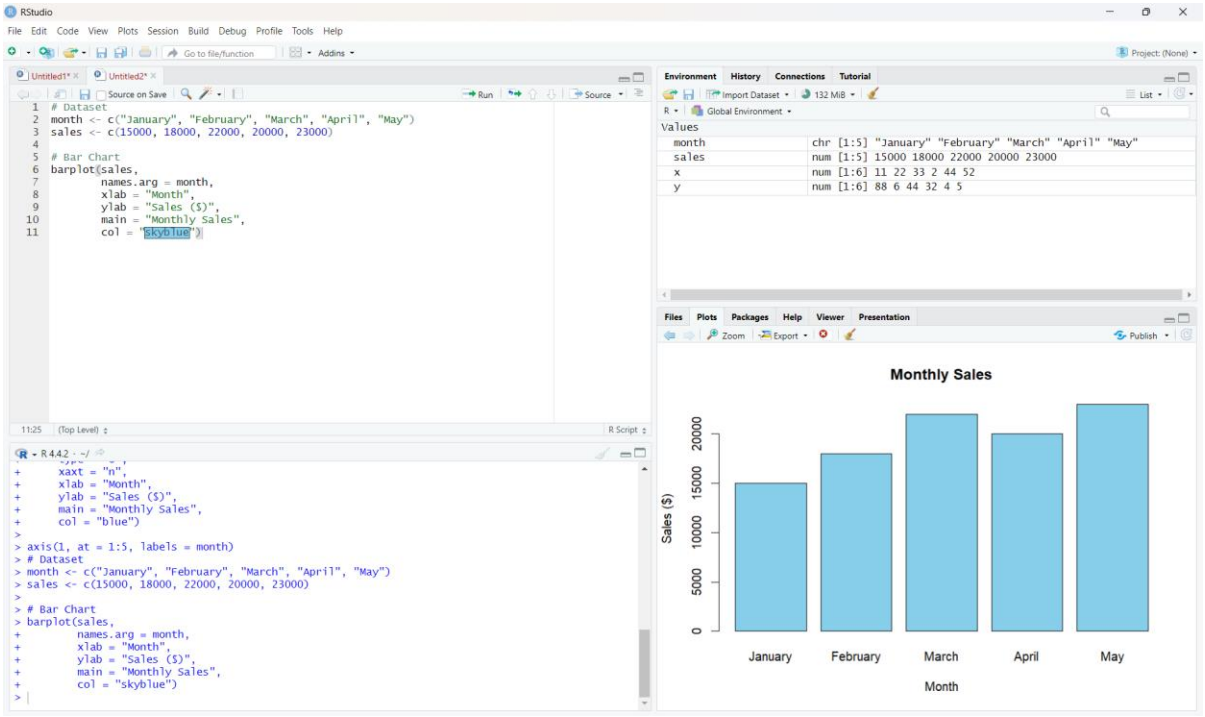
Observation: Sales increase from January to March, decrease slightly in April, and reach the highest value in May.

PROGRAM 2: Bar Chart of Sales

Generate a bar chart to display the sales values for the available months and label the chart elements.

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

# Bar Chart
barplot(sales,
        names.arg = month,
        xlab = "Month",
        ylab = "Sales ($)",
        main = "Monthly Sales",
        col = "skyblue")
```



Output 2: Monthly Sales Bar Chart

Observation: The bar chart makes it easy to compare the sales of each month. May has the highest sales and January has the lowest.

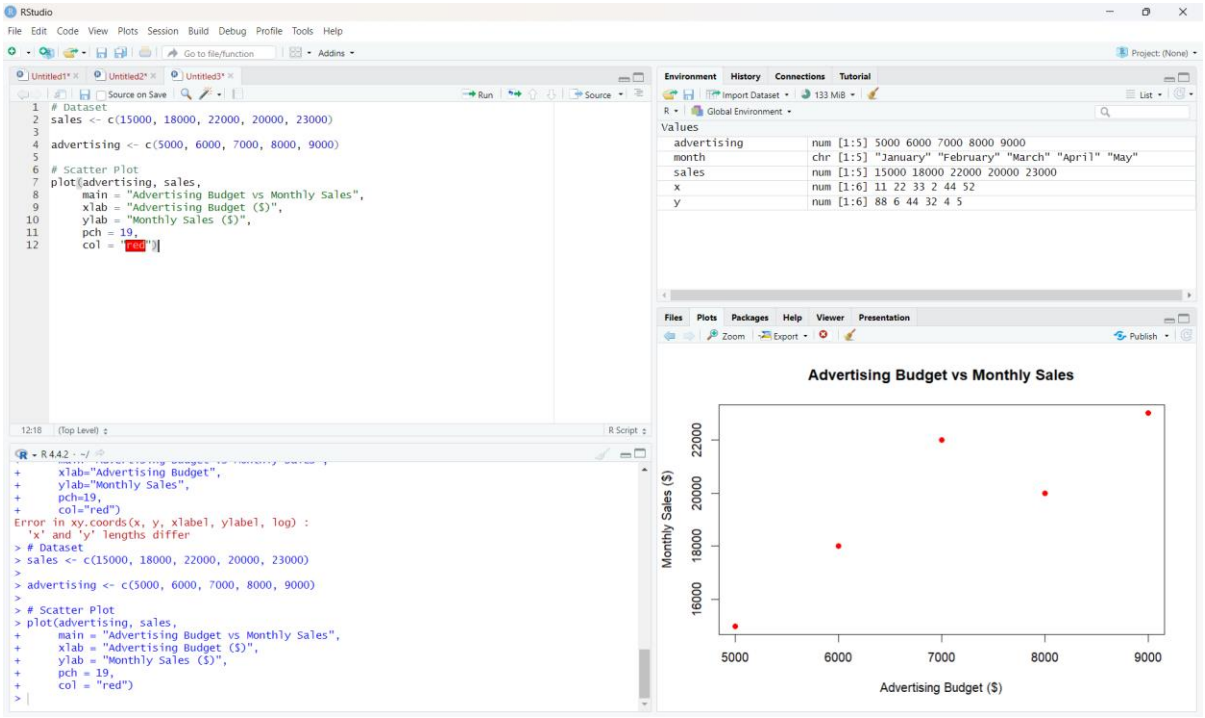
PROGRAM 3: Scatter Plot of Advertising Budget vs Monthly Sales

Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

```
# Dataset
sales <- c(15000, 18000, 22000, 20000, 23000)

advertising <- c(5000, 6000, 7000, 8000, 9000)

# Scatter Plot
plot(advertising, sales,
     main = "Advertising Budget vs Monthly Sales",
     xlab = "Advertising Budget ($)",
     ylab = "Monthly Sales ($)",
     pch = 19,
     col = "red")
```



Output 3: Advertising Budget vs Monthly Sales Scatter Plot

Insight: The plotted points show a generally positive relationship: as the advertising budget increases from $5,000 to $9,000, monthly sales also generally increase. This suggests that higher advertising spending is associated with higher sales for the values plotted.

Note: The visible assignment table contains only Month and Sales. The advertising values used in the submitted R program are the five values shown in the user's executed program: 5000, 6000, 7000, 8000, and 9000.

PROGRAM 4: Interactive Sales Dashboard

Build an interactive dashboard combining the line chart and bar chart using the Shiny package.

```
library(shiny)

month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

ui <- fluidPage(

  titlePanel("Monthly Sales Dashboard"),

  sidebarLayout(

    sidebarPanel(
      selectInput("chart",
                  "Select Chart:",
                  choices = c("Line Chart", "Bar Chart"))
    ),

    mainPanel(
      plotOutput("salesPlot")
    )
  )
)

server <- function(input, output) {

  output$salesPlot <- renderPlot({

    if (input$chart == "Line Chart") {

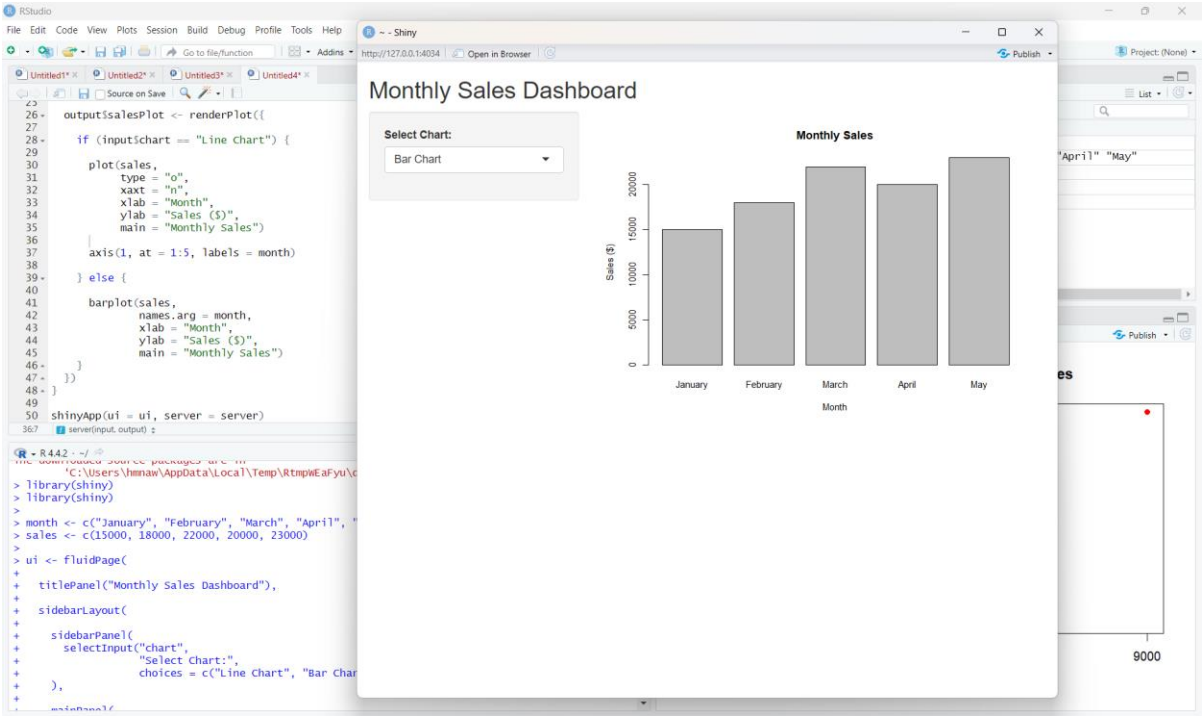
      plot(sales,
           type = "o",
           xaxt = "n",
           xlab = "Month",
           ylab = "Sales ($)",
           main = "Monthly Sales")

      axis(1, at = 1:5, labels = month)

    } else {

      barplot(sales,
              names.arg = month,
              xlab = "Month",
              ylab = "Sales ($)",
              main = "Monthly Sales")
    }
  })
}

shinyApp(ui = ui, server = server)
```



Output 4: Interactive Monthly Sales Dashboard

Observation: The dashboard provides a chart-selection control. Selecting Line Chart displays the monthly trend, while selecting Bar Chart displays the monthly sales comparison.

Procedure

1. Create the required month and sales vectors in R.

2. Use plot() to create the line chart with appropriate axis labels and title.

3. Use barplot() to compare the sales values for the available months.

4. Create advertising-budget and sales vectors of equal length and use plot() for the scatter plot.

5. Install and load the Shiny package, then create the UI and server components for the interactive dashboard.

6. Run the Shiny application and switch between the Line Chart and Bar Chart using the selection box.

Result

The monthly sales data was successfully visualized using line, bar, and scatter charts. An interactive Shiny dashboard was also created to switch between the line chart and bar chart.

Key R Functions Used

Function	Purpose
plot()	Creates line and scatter plots.
axis()	Adds custom axis labels.
barplot()	Creates bar charts.
fluidPage()	Creates the Shiny page layout.
selectInput()	Provides the chart selection control.
renderPlot()	Generates plots dynamically in Shiny.
shinyApp()	Runs the interactive Shiny application.

SET 2: CUSTOMER FEEDBACK ANALYSIS

Dataset: Customer Satisfaction

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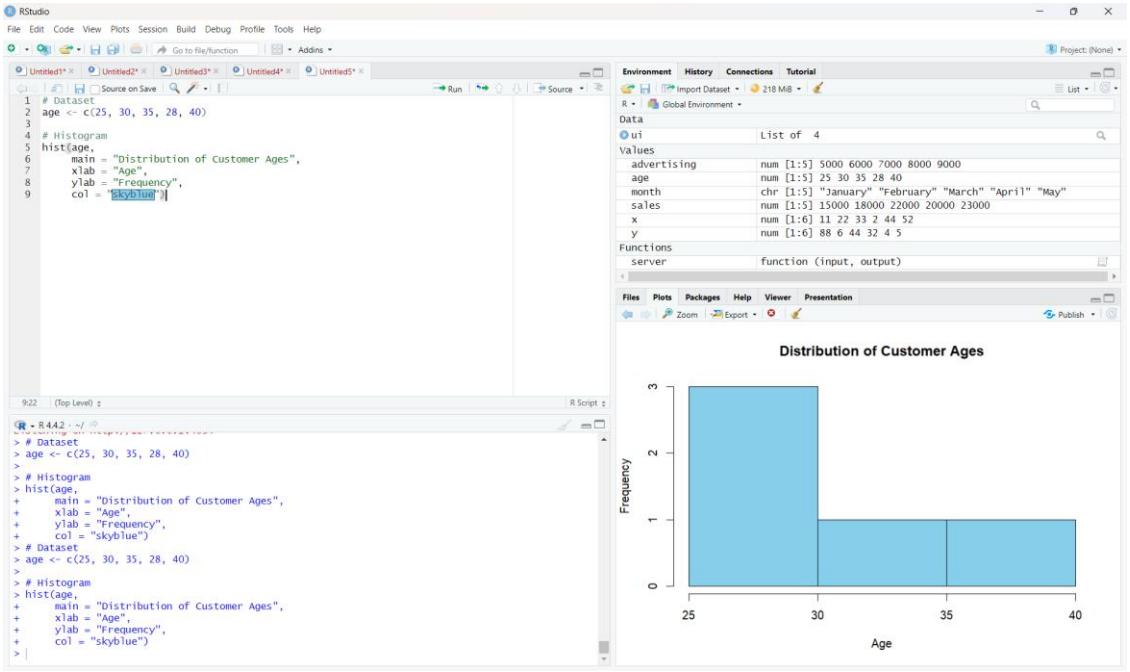
Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1: Histogram of Customer Ages

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
# Dataset
age <- c(25, 30, 35, 28, 40)

# Histogram
hist(age,
     main = "Distribution of Customer Ages",
     xlab = "Age",
     ylab = "Frequency",
     col = "skyblue")
```



Output 1: Distribution of Customer Ages

Observation: The histogram represents the distribution of the five customer ages in the given dataset.

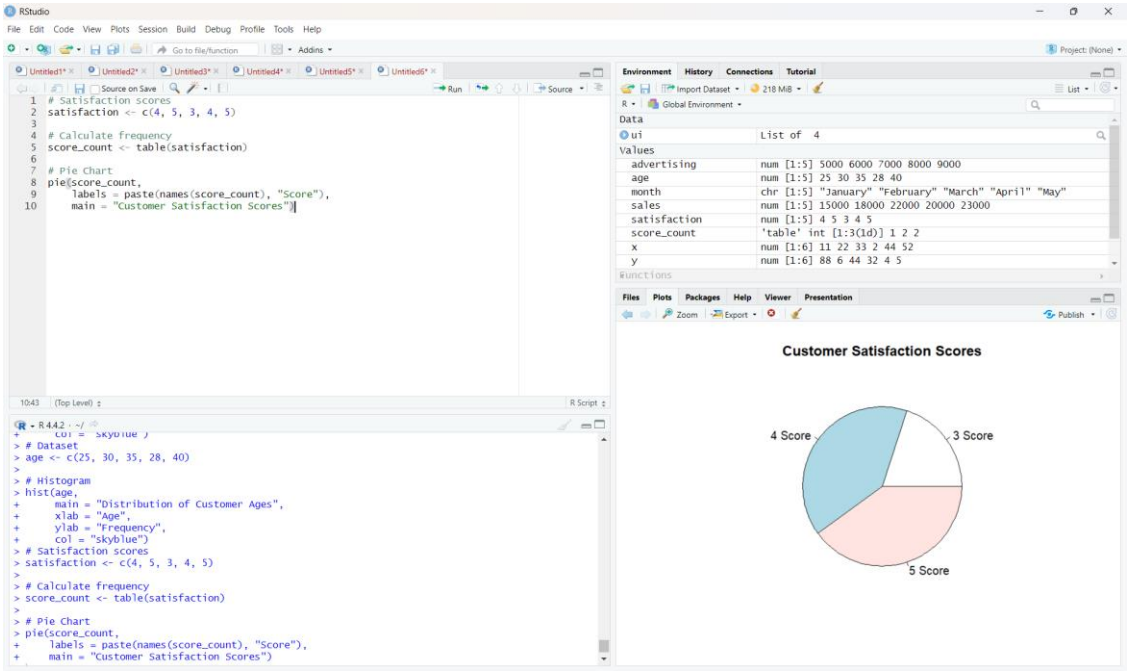
PROGRAM 2: Pie Chart of Satisfaction Scores

Generate a pie chart to display the overall distribution of customer satisfaction scores and include labels.

```
# Satisfaction scores
satisfaction <- c(4, 5, 3, 4, 5)

# Calculate frequency
score_count <- table(satisfaction)

# Pie Chart
pie(score_count,
    labels = paste(names(score_count), "Score"),
    main = "Customer Satisfaction Scores")
```



Output 2: Customer Satisfaction Scores Pie Chart

Observation: Scores 4 and 5 each occur twice, while score 3 occurs once.

PROGRAM 3: Stacked Bar Chart by Age Group

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

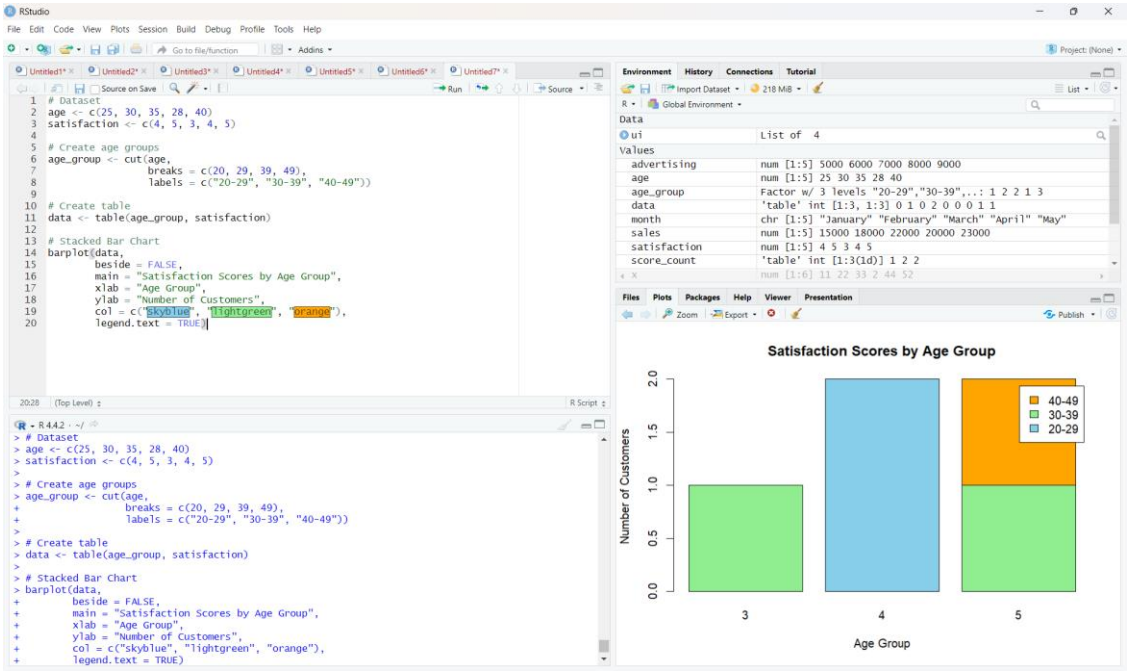
```
# Dataset
age <- c(25, 30, 35, 28, 40)
satisfaction <- c(4, 5, 3, 4, 5)

# Create age groups
age_group <- cut(age,
                 breaks = c(20, 29, 39, 49),
                 labels = c("20-29", "30-39", "40-49"))

# Create table
data <- table(age_group, satisfaction)

# Stacked Bar Chart
barplot(data,
        beside = FALSE,
```

```
        main = "Satisfaction Scores by Age Group",
        xlab = "Age Group",
        ylab = "Number of Customers",
        col = c("skyblue", "lightgreen", "orange"),
        legend.text = TRUE)
```



Output 3: Satisfaction Scores by Age Group

Observation: The code creates three age groups and stacks the satisfaction-score frequencies for each group.

Note: The supplied RStudio screenshot is inserted exactly as captured, including its displayed x-axis labels.

PROGRAM 4: Word Cloud of Customer Feedback

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```
library(wordcloud)
library(RColorBrewer)

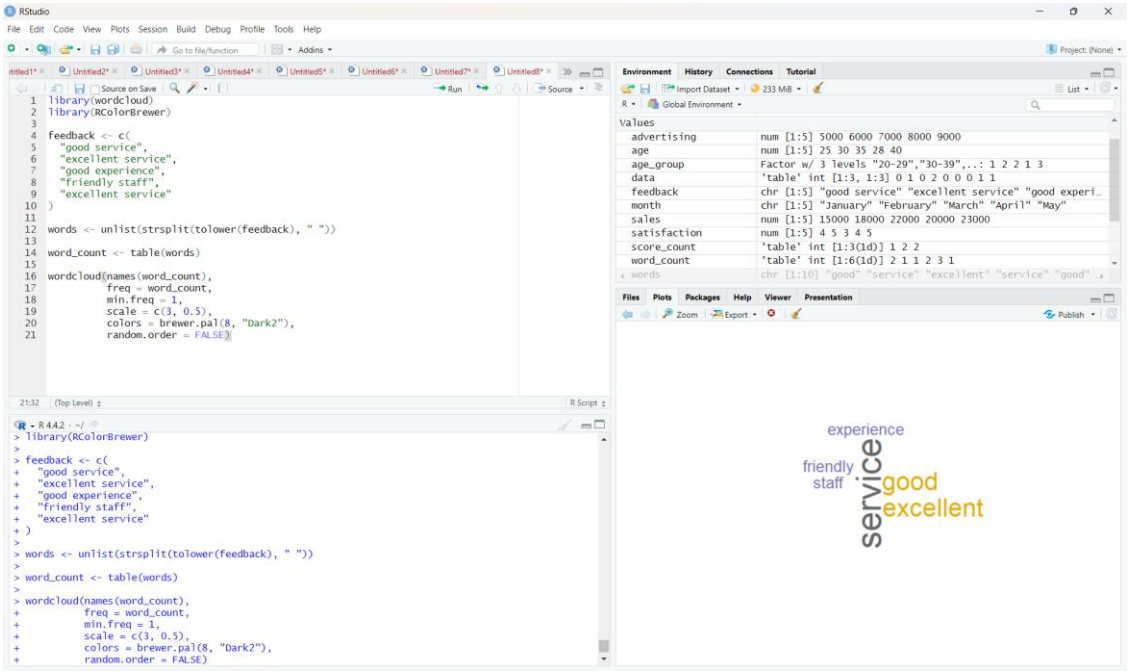
feedback <- c(
  "good service",
  "excellent service",
  "good experience",
  "friendly staff",
  "excellent service"
)

words <- unlist(strsplit(tolower(feedback), " "))

word_count <- table(words)

wordcloud(names(word_count),
          freq = word_count,
          min.freq = 1,
          scale = c(3, 0.5),
          colors = brewer.pal(8, "Dark2"),
```

```
        random.order = FALSE)
```



Output 4: Customer Feedback Word Cloud

Observation: The word cloud highlights frequently occurring terms in the supplied sample feedback, including service, excellent, good, friendly, and experience.

Set 2 Procedure

1. Create vectors for customer age and satisfaction score.

2. Use hist() to represent the distribution of customer ages.

3. Use table() and pie() to display the satisfaction score distribution.

4. Create age groups using cut(), form a frequency table, and use barplot() for the stacked bar chart.

5. Prepare open-ended feedback text, count the words, and use wordcloud() to visualize frequent terms.

Set 2 Result

The customer satisfaction data was successfully visualized using a histogram, pie chart, stacked bar chart, and word cloud.

SET 3: EMPLOYEE PERFORMANCE EVALUATION

Aim

To visualize monthly sales and employee performance data using line, bar, and scatter charts, analyze the relationship between years of service and performance scores, and outline the creation of an interactive Tableau dashboard.

Dataset 1: Monthly Sales

Month	Sales ($)
January	15000
February	18000
March	22000
April	20000
May	23000

Dataset 2: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

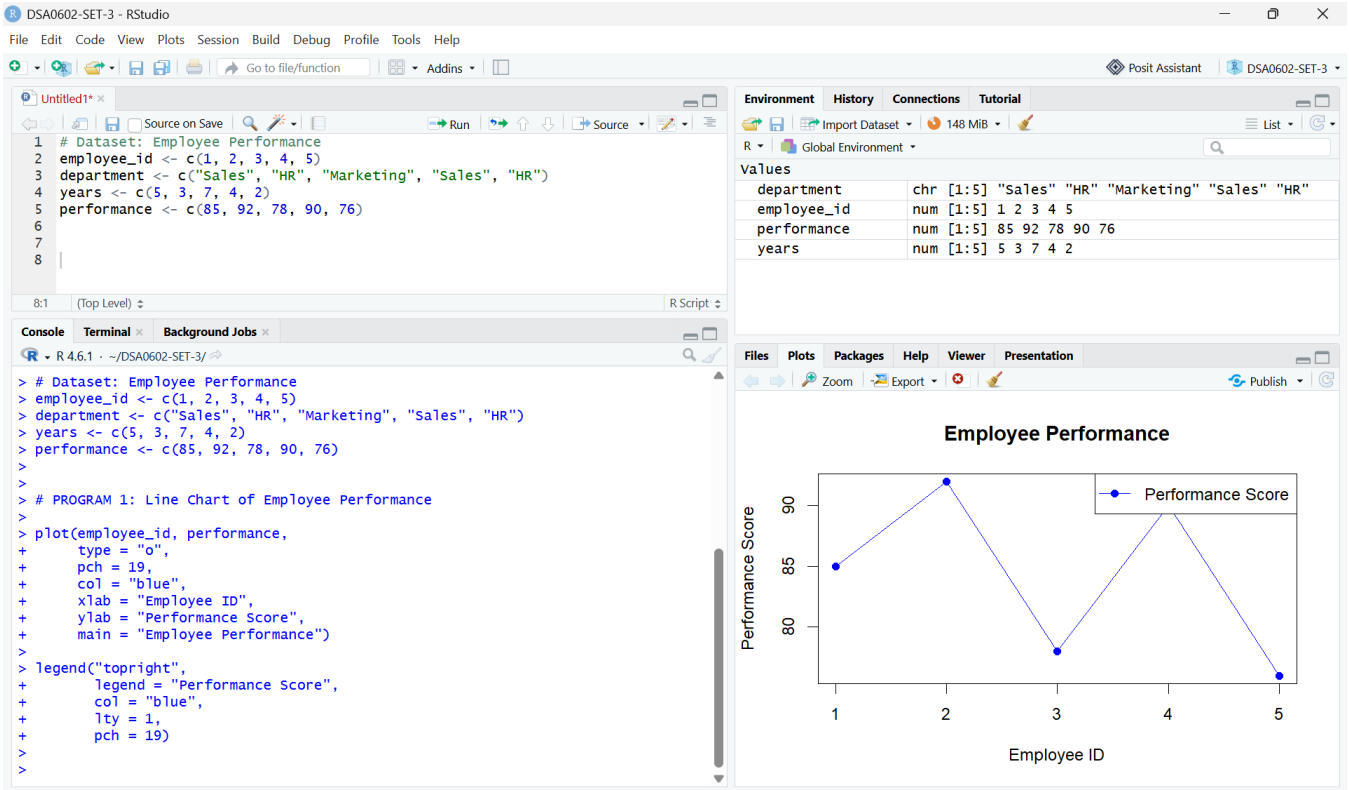
PROGRAM 1: Line Chart of Employee Performance

The employee dataset contains one score per employee but no date or month field. Therefore, the employee performance line chart uses Employee ID on the x-axis. The separate monthly sales chart shows the available time-based trend.

```
employee_id <- c(1, 2, 3, 4, 5)
performance <- c(85, 92, 78, 90, 76)

plot(employee_id, performance, type = "o", pch = 19,
     col = "blue", xlab = "Employee ID",
     ylab = "Performance Score",
     main = "Employee Performance Scores")
legend("topright", legend = "Performance Score",
       col = "blue", lty = 1, pch = 19)

month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
plot(sales, type = "o", pch = 19, col = "darkgreen",
     xaxt = "n", xlab = "Month", ylab = "Sales ($)",
     main = "Monthly Sales Trend")
axis(1, at = 1:5, labels = month)
legend("topleft", legend = "Monthly Sales",
       col = "darkgreen", lty = 1, pch = 19)
```



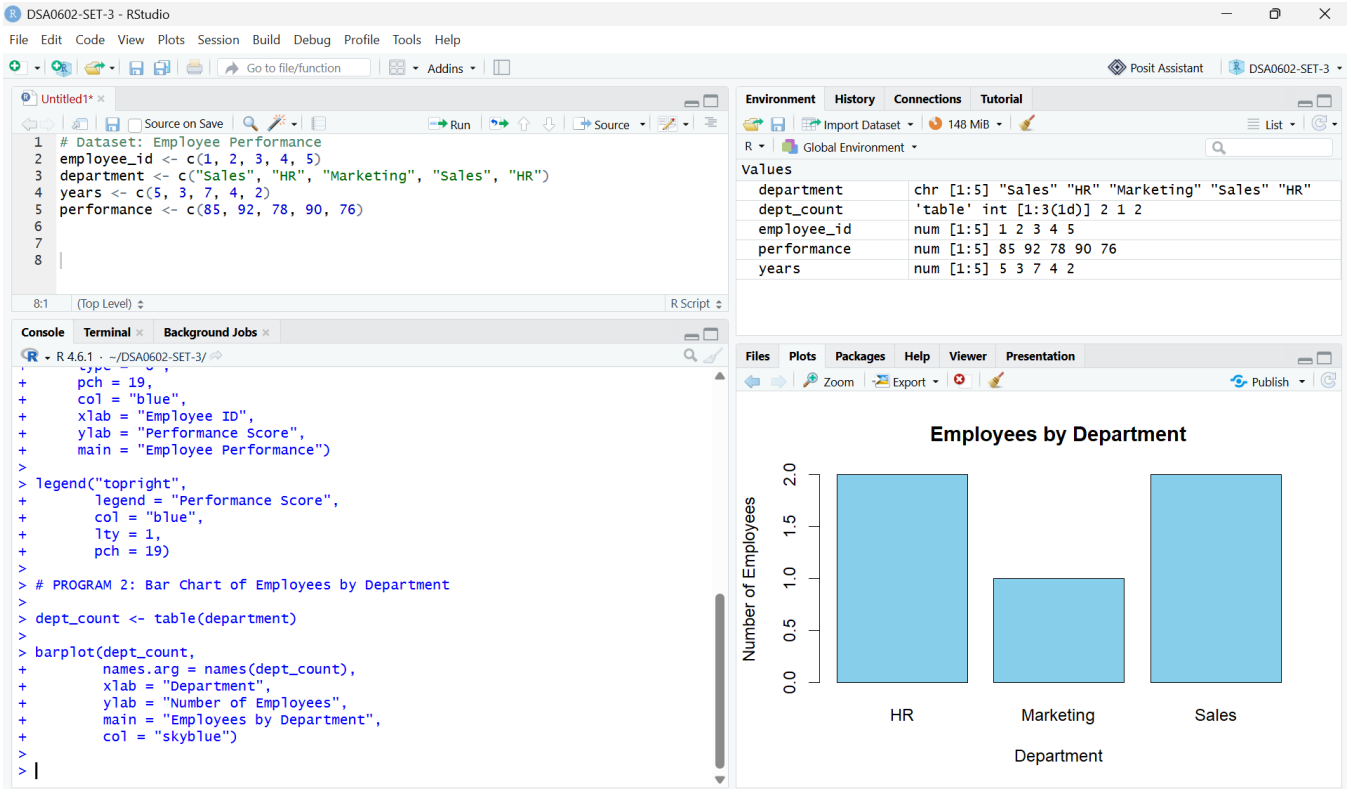
Output 1A: Employee Performance Scores by Employee ID

Observation: Employee 2 has the highest score (92), while Employee 5 has the lowest (76). This is a comparison across employees, not a time trend.

PROGRAM 2: Bar Chart of Employees by Department

Generate a bar chart showing the number of employees in each department and label the chart elements.

```
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
dept_count <- table(department)
barplot(dept_count, names.arg = names(dept_count),
        xlab = "Department", ylab = "Number of Employees",
        main = "Employees by Department",
        col = "skyblue", ylim = c(0, 3))
```



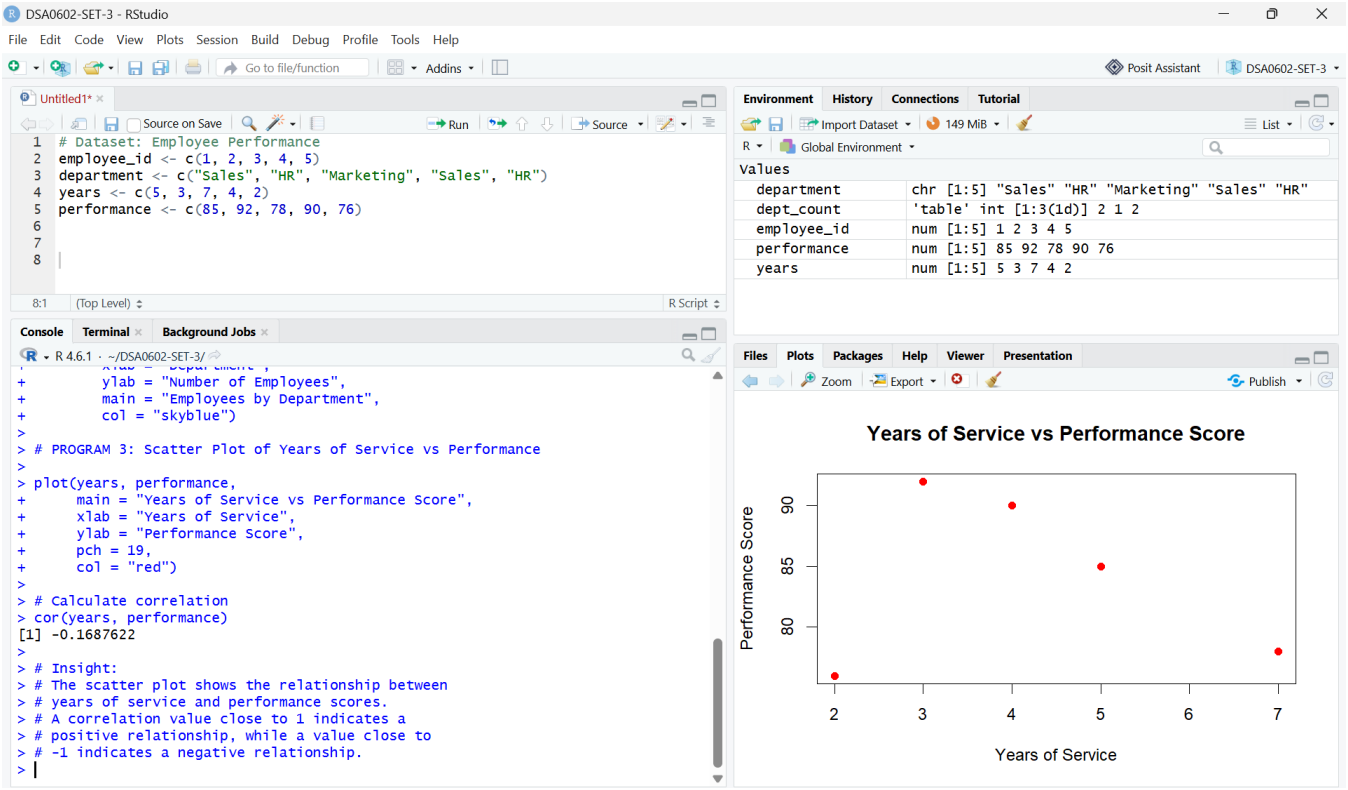
Output 2: Employees by Department

Observation: Sales and HR each contain 2 employees, while Marketing contains 1 employee.

PROGRAM 3: Scatter Plot of Years of Service vs Performance

Create a scatter plot to examine the relationship between years of service and performance score.

```
years <- c(5, 3, 7, 4, 2)
performance <- c(85, 92, 78, 90, 76)
plot(years, performance,
     main = "Years of Service vs Performance Score",
     xlab = "Years of Service",
     ylab = "Performance Score",
     pch = 19, col = "red")
text(years, performance, labels = 1:5, pos = 3, cex = 0.8)
cor(years, performance, method = "pearson")
```



Output 3: Years of Service vs Performance Score

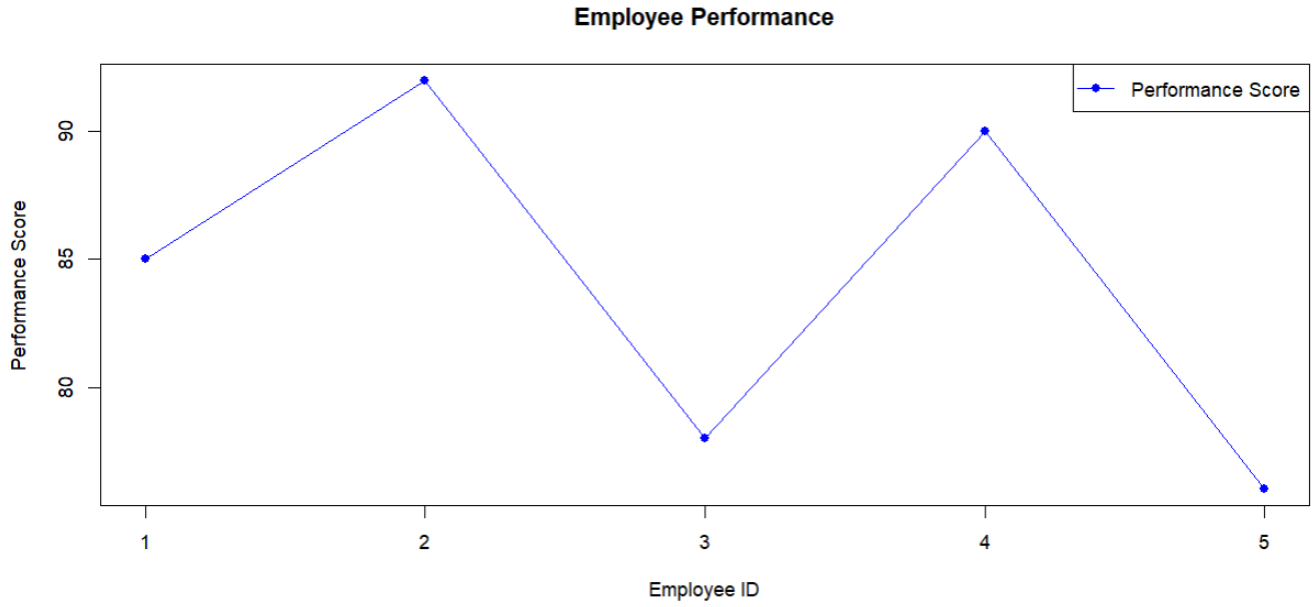
Insight: The Pearson correlation for these five records is approximately -0.169, showing a negative linear association in this small sample. Employees with more years of service do not necessarily have higher scores in these data. Correlation does not establish causation.

PROGRAM 4: Tableau Dashboard

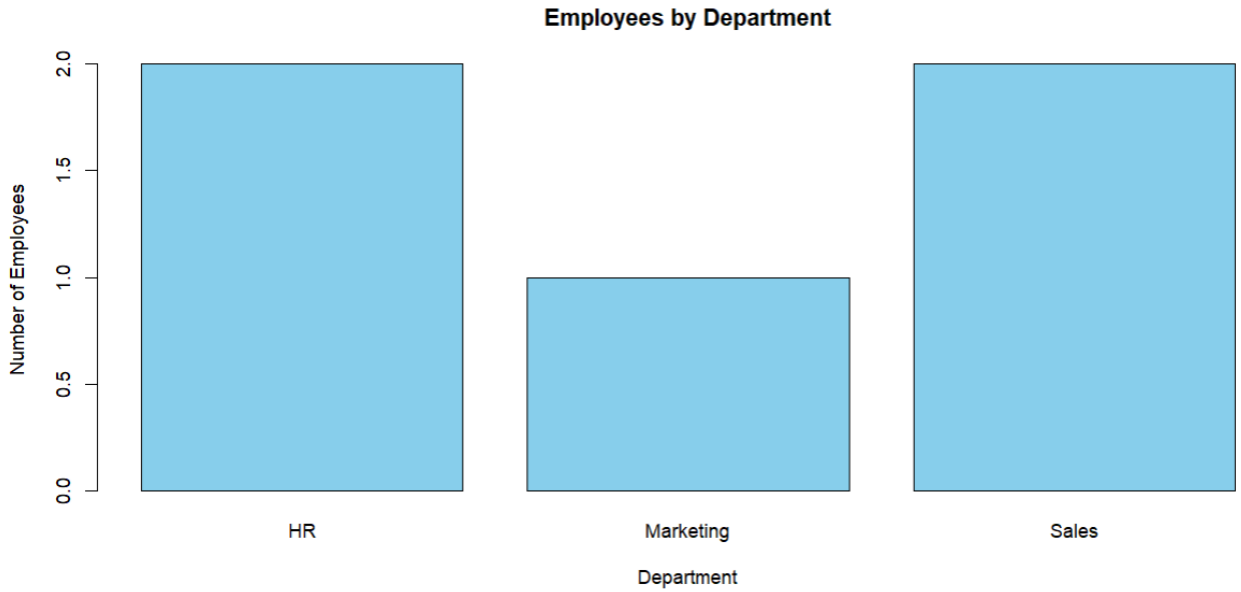
Create an interactive Tableau dashboard containing the monthly sales line chart and department-wise employee bar chart.

Procedure in Tableau

1. Open Tableau Desktop or Tableau Public and connect to the data source. Import or enter the Monthly Sales and Employee Performance tables.
2. Create a worksheet named Monthly Sales Trend. Place Month on Columns and Sales on Rows, then select Line in the Marks card. Add a descriptive title.
3. Create a worksheet named Employees by Department. Place Department on Columns and Employee ID on Rows; change the aggregation to COUNT, and select Bar.
4. Create a new Dashboard and drag both worksheets onto the dashboard canvas.
5. Add Month and Department filters where appropriate. Configure filters to apply to the relevant worksheets.
6. Enable tooltips and optionally add dashboard filter actions to explore related views.
7. Save the workbook as a Tableau packaged workbook (.twbx) or publish it using the available Tableau option.



Output 4: Suggested Tableau Dashboard Layout (illustrative)



Expected behavior: The dashboard displays the monthly sales trend and department distribution together. Filters and tooltips support interactive exploration after the workbook is built in Tableau.

Set 3 Procedure

8. Enter the monthly sales and employee performance values in R as vectors.
9. Use plot() to draw employee scores against Employee ID and include axis labels, a title, and a legend.
10. Use plot() with month labels to draw the monthly sales line chart.
11. Use table() to count employees by department and barplot() to visualize the counts.

12. Use plot() for the years-of-service versus performance scatter plot and cor() to calculate Pearson correlation.
13. In Tableau, create the two worksheets, assemble them on a dashboard, and add filters and tooltips.

Result

The monthly sales and employee performance data were visualized using line charts, a department-wise bar chart, and a scatter plot. The department distribution and association between years of service and performance scores were examined. The steps to create an interactive Tableau dashboard with a monthly sales line chart and department bar chart were documented. A true employee performance trend over time requires dated performance observations for each employee.

Key R Functions Used

Function	Purpose
plot()	Creates line and scatter plots.
axis()	Adds custom axis tick labels.
legend()	Adds a chart legend.
barplot()	Creates a bar chart.
table()	Calculates frequency counts.
cor()	Calculates the correlation coefficient.